

Statins are a group of drugs that help in reducing blood cholesterol levels

Protein

Adequate protein intake from lean meat, egg white, pulses, small oily fish rich in omega 3 fatty acids, low - fat milk, nuts and whole grains should be provided. Legumes, beans, peas, and lentils are good sources of plant protein that contain less fat, making them good substitutes for meat. Substituting plant protein for animal protein will reduce cholesterol intake and increase fibre intake.

Vitamins and minerals

Vitamins and minerals play an important role in maintaining good health. In CVD, sodium intake should be restricted by restricting salt intake. Low - fat dairy can be included in the diet to ensure adequate calcium intake. Inclusion of foods rich in B vitamins such as folic acid, B6 and B12 reduces risk for CVD.

11.13 Cardio protective effect of functional foods

The National Academy of Sciences (1994) defines functional foods as "any food or food ingredient that may provide a health benefit beyond the traditional nutrients it contains".

The most common functional foods that have a cardio-protective effect include omega-3 fatty acids in fish and nuts, dietary fibre, and phytochemicals present in vegetables particularly soy. Some of the functional foods, their potential mechanism of action and the bioactive compounds responsible for it are tabulated in table 11.3.

Table 11.3 : Possible cardio-protective effects of functional foods

Functional foods	Bioactive compound	Potential mechanism
Nuts (walnuts, almond)	Tocopherols, omega-3 fatty Acids	Lowers blood cholesterol
Legumes	Fibre and polyphenols	
Garlic	Allicin	
Whole grains	Fibre and phytochemicals	
Dark chocolate	Flavonoid	
Small oily fish	Omega-3 fatty acids	• Lowers blood cholesterol • Inhibits of LDL-C oxidation • Lowers blood triglycerides
Citrus fruits and vegetables	Vitamin C	• Inhibits of LDL-C oxidation • Antioxidant action • Lowers blood pressure
Green tea and black tea	Polyphenols	
Tomato	Lycopene	
Soy proteins	Genistein and Daidzein	• Lowers blood cholesterol • Inhibits of LDL-C oxidation • Antioxidant action

Source: Eman M. Alissa and Gordon A. Ferns (2012), Functional Foods and Nutraceuticals in the Primary Prevention of Cardiovascular Diseases, Journal of Nutrition and Metabolism, pg 4,5.

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